

GAGAN SINGHAL

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PROFESSIONAL SUMMARY

AI builder and storyteller at **Boston University**, blending code, creativity, and curiosity to craft meaningful AI systems. Skilled with transformers, vision models, and data-driven systems. With **4 years** of experience across analytics, research, and experimentation, I'm chasing **2026 full-time** roles in AI/ML dev or research. Let's build something intelligent that learns and lasts.

EDUCATION

Boston University, Massachusetts, USA

M.S. in Artificial Intelligence

September, 2024 - December, 2025

GPA: 3.87/4

Thapar University, Punjab, India

Bachelor's in Computer Science

July, 2018 - June, 2022

GPA: 3.6/4

SKILLS

Programming Languages: Python, Java, C++, SQL, JavaScript

Frameworks/Libraries: PyTorch, TensorFlow, HuggingFace, Scikit-learn, OpenCV, NumPy, Pandas, Node.js

Core AI/ML Skills: Multi-Modal ML, LangChain, LangGraph, RAG, Computer Vision, NLP, Transformers, LLM Fine-tuning

Engineering Skills: Git, Oracle DB, REST APIs, Cloud Deployment, Agile Development

Soft Skills: Teamwork, Troubleshooting, Clear Communication, Strategic Thinking

EXPERIENCE

Oracle Corporation - Software Developer I

Aug, 2022 - Nov, 2023

- **Led a team of 3** to spearhead R&D initiatives embedding machine learning into **enterprise-grade forecasting, profit optimization, and analytics workflows**.
- Developed **Java-based automation frameworks**, saving time of development cycles by **3-5 man days** through intelligent testing.

Oracle Corporation - Software Developer Intern

Jan, 2022 - Jul, 2022

- **Owned** and built **ML-driven analytics foundations** for enterprise software.
- Architected distributed-performance monitoring with **Zipkin, Prometheus, and Java EE**.

PROJECTS

WatchIt – Autonomous Screen Monitoring Agentic AI (LangGraph, OpenAI, Postgres)

Sept, 2025 – Now

- Engineered an **AI agent** that autonomously classifies on-screen content for **real-time parental control and content moderation**.
- Built a modular **LangGraph multi-agent system** integrating **LLM-based vision, reasoning, and policy nodes** with a **PostgreSQL + vector database backend** for contextual memory and adaptive rule learning.
- Deployed with a dashboard for content review and **admin settings**, reducing manual intervention by **70% to 80%**.

AlphaBeat - Music Generation using AI (PyTorch, Latent Diffusion, AR Transformers)

Feb, 2025 - May, 2025

- Designed and benchmarked **8 hybrid pipelines** combining **transformer and latent diffusion models** to enhance **text-to-music generation using melody conditioning**.
- Demonstrated that cascaded architectures significantly improve both **semantic alignment (CLAP↑ 0.35)** and **audio fidelity (FAD↓ 6.55)** over standalone baselines, with results validated via listening tests and quantitative metrics.

Data-Driven Transit Analysis & Forecasting (Pandas, Plotly, XGBoost, Random Forest)

Mar, 2025 - April, 2025

- Conducted in-depth data analysis on **8 years of MBTA bus system data (2015-2024)** to evaluate **ridership trends, service reliability, and transit equity**.
- Integrated machine learning models (**XGBoost, Random Forest**) for predicting bus on-time performance and ridership volume, enabling targeted scheduling improvements.
- Created **interactive visualizations** to communicate **actionable insights** and **presented data-driven policy recommendations**.

Sign Language Translation (PyTorch, DenseNet, LSTM)

Oct, 2024 - Dec, 2024

- Built **real-time translation system** for sign language using **DenseNet + LSTM** for spatio-temporal feature modeling.
- Trained on **WLASL dataset**; achieved high-accuracy gesture-to-text translation with optimized preprocessing and augmentation.

Notable Mentions

2021

- **Drowsiness Detection System (DLib, CNN)** (Apr'21)
- **NLP COVID Information Tool (Transformers)** (Jan'21 - Mar'21)